10) Creation of VHD on Windows and restoring on different host

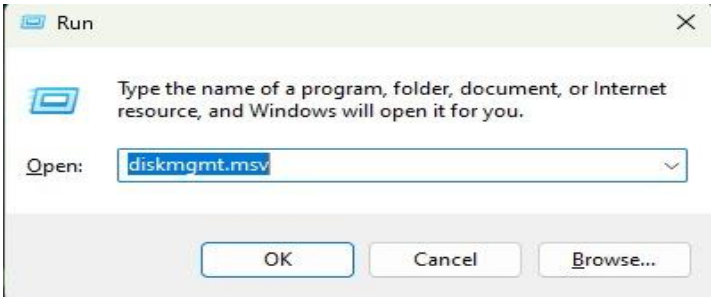
Aim:

To create a VHD on windows and restore the same in different host

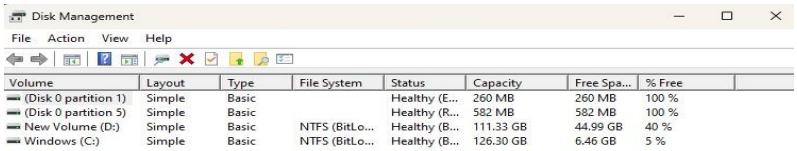
Steps:

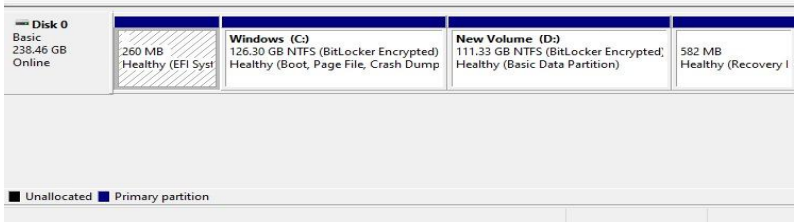
<u>Create a VHD on Windows</u>

1. Press Win + R, type diskmgmt.msc, and press Enter



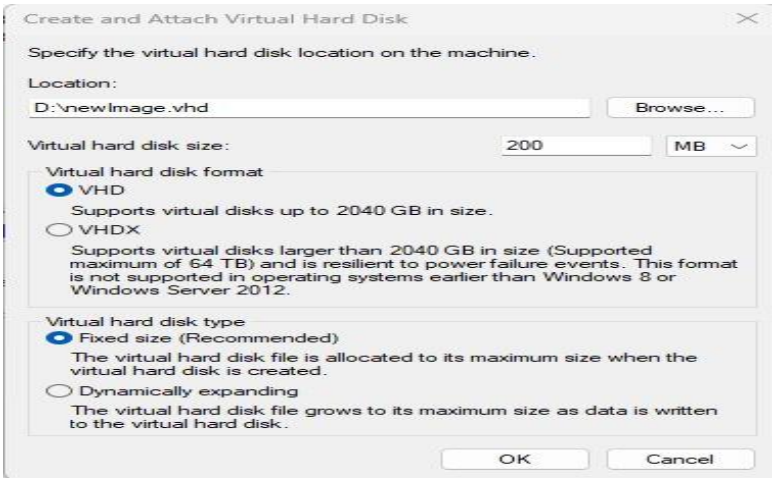
2. In Disk Management, go to the Action menu → Create VHD





3. Location → Browse and select where to save the .vhd or .vhdx file.
4. Virtual Hard Disk Size (decide based on what you need).
5. VHD format:VHD (compatible with older systems, max 2TB).

6. Type: Fixed size → Pre-allocates the full size. Faster runtime, but uses disk space upfront.
7. Dynamically expanding → Grows as data is added. Saves space initially.
8. Click OK → The VHD will be created and appear in Disk Management as "Not Initialized".

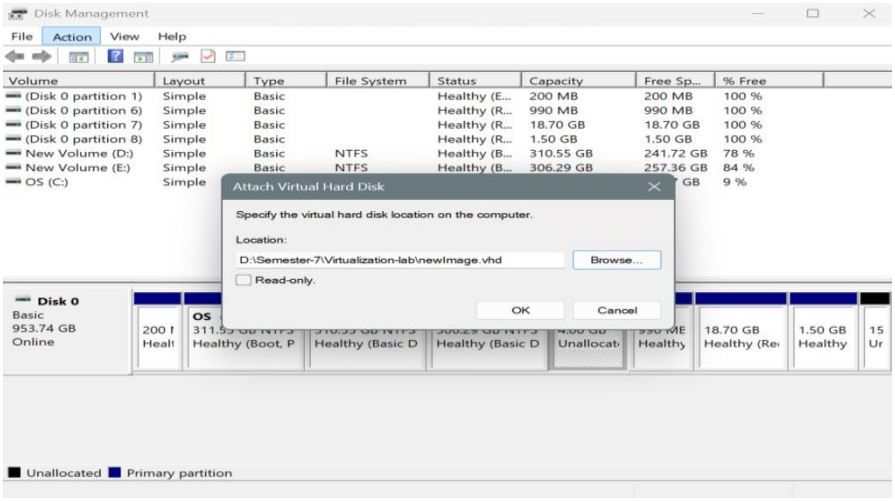


9. Right-click the new disk → Initialize Disk.
10. Create a new volume: Right-click unallocated space → New Simple Volume → Assign a drive letter → Format (NTFS recommended).
11. Now you can use it like a normal drive.

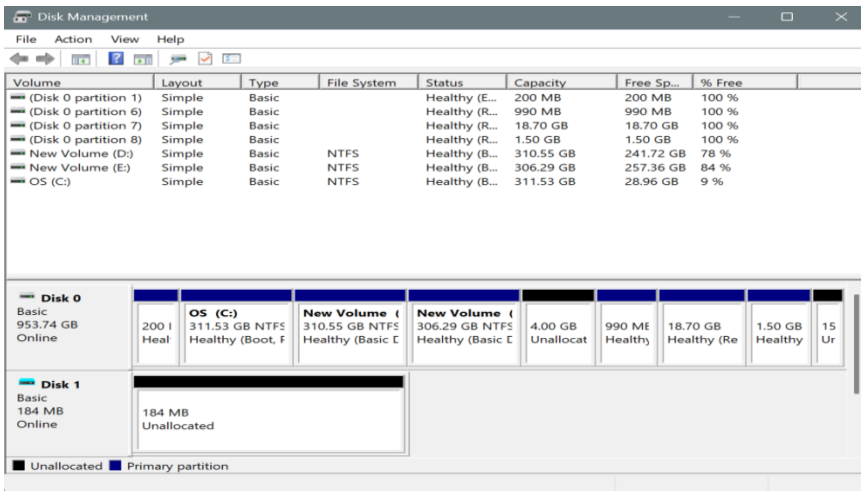


Steps to Use on Another Host

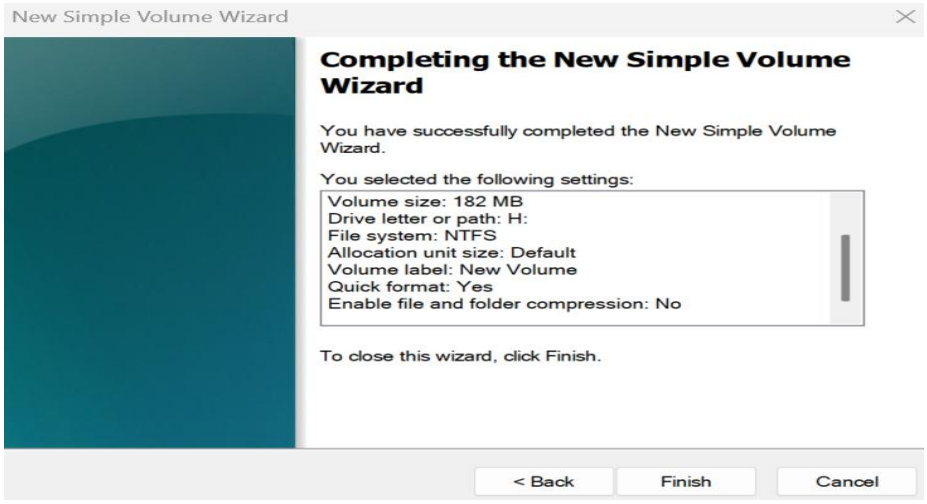
1. Copy your .vhd or .vhdx file to the new machine (via external HDD, network, or cloud).
2. On the new host:Open Disk Management → Action → Attach VHD.
3. Browse to the .vhd file and click OK.



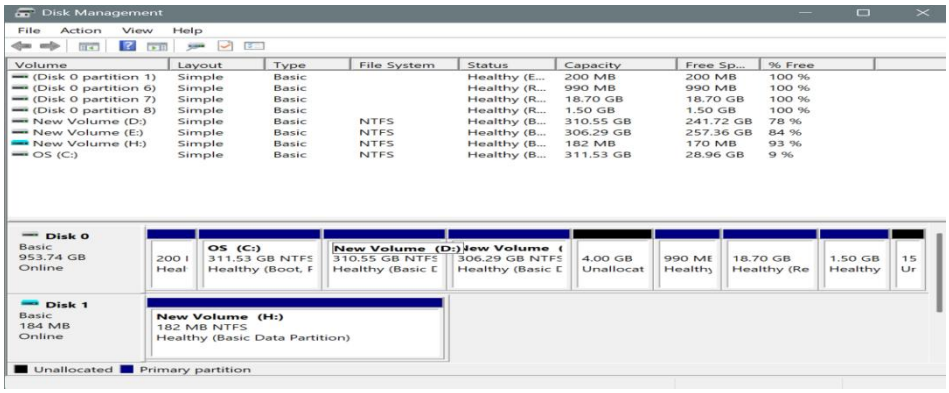
4. The VHD will appear as a new drive (with the same partitions/data).



5. If needed, assign a drive letter:Right-click the partition → Change Drive Letter and Paths → Assign a letter.



6. Now you can access files just like a normal disk.

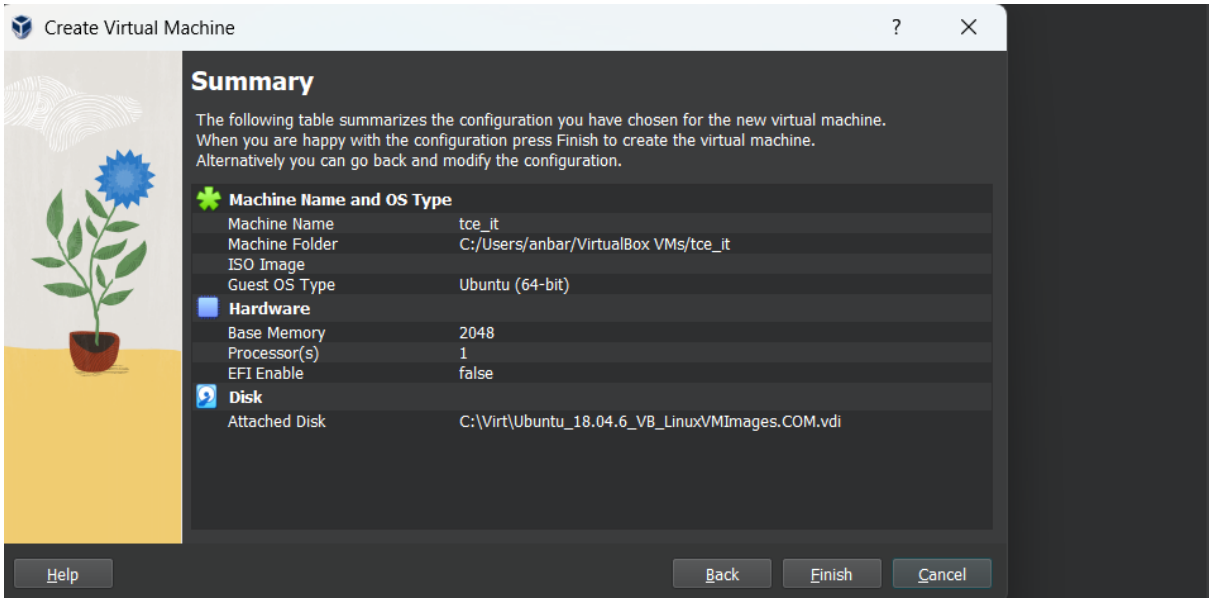


STORAGE VIRTUALIZATION:

Storage Virtualization

Step 1: Create a VM and Add Additional Storage

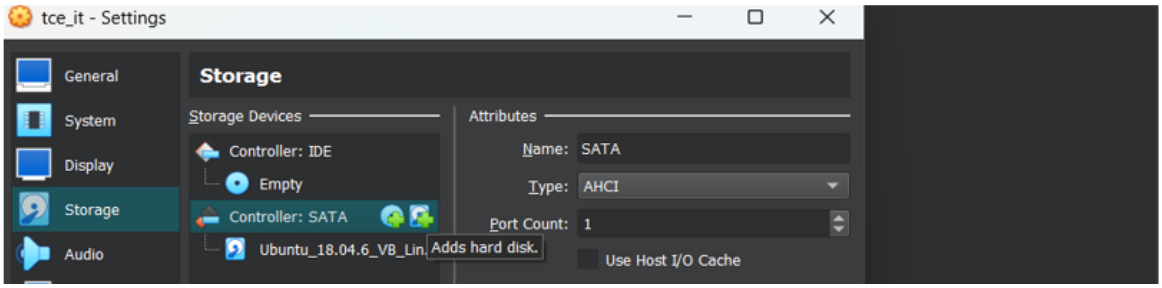
- **Create a Linux VM**
  - **Open VirtualBox → Click New → Enter VM Name, Type = Linux, Version = ubuntu.**
  - **Allocate RAM (e.g., 2–4 GB).**
  - **Hard Disk: Select "Use an existing virtual hard disk file".**
  - **Browse and select your VDI file.**

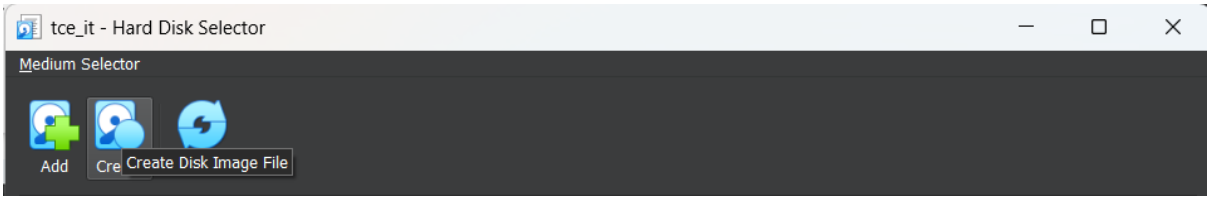


  - **Click Finish.**

- **Add an Additional Virtual Disk**
  - **This disk will be used for extending storage via LVM.**
  - **In VirtualBox:**

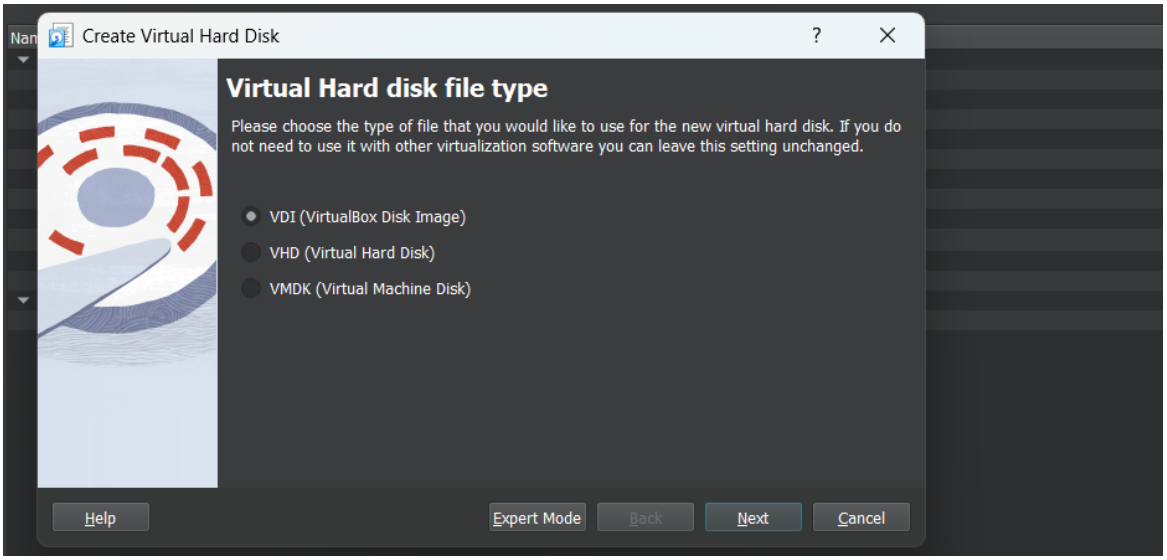
1. **Select the VM → Settings → Storage.**

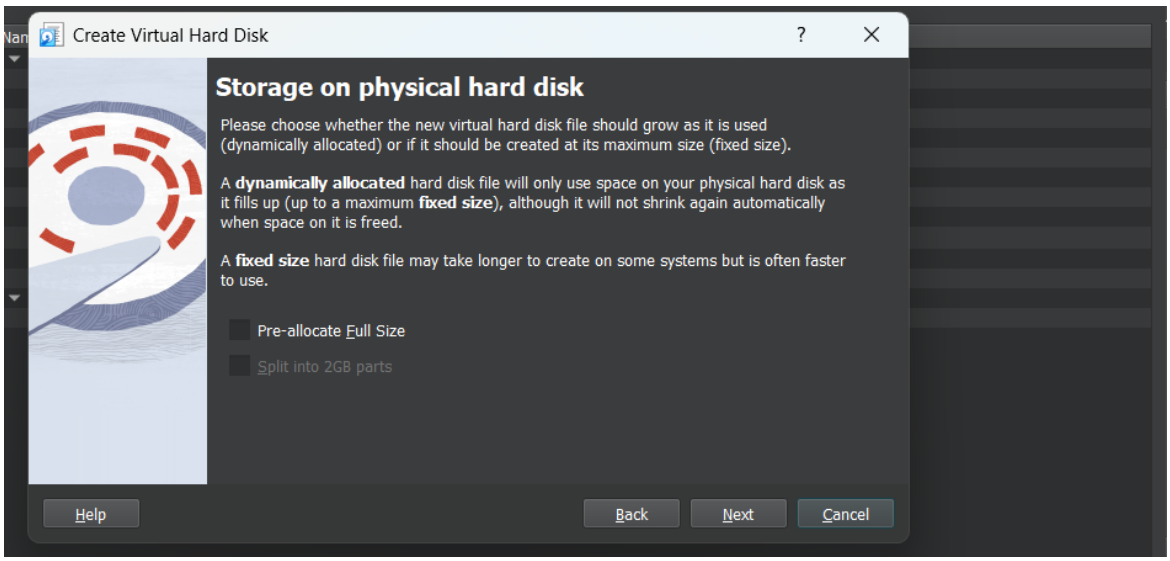




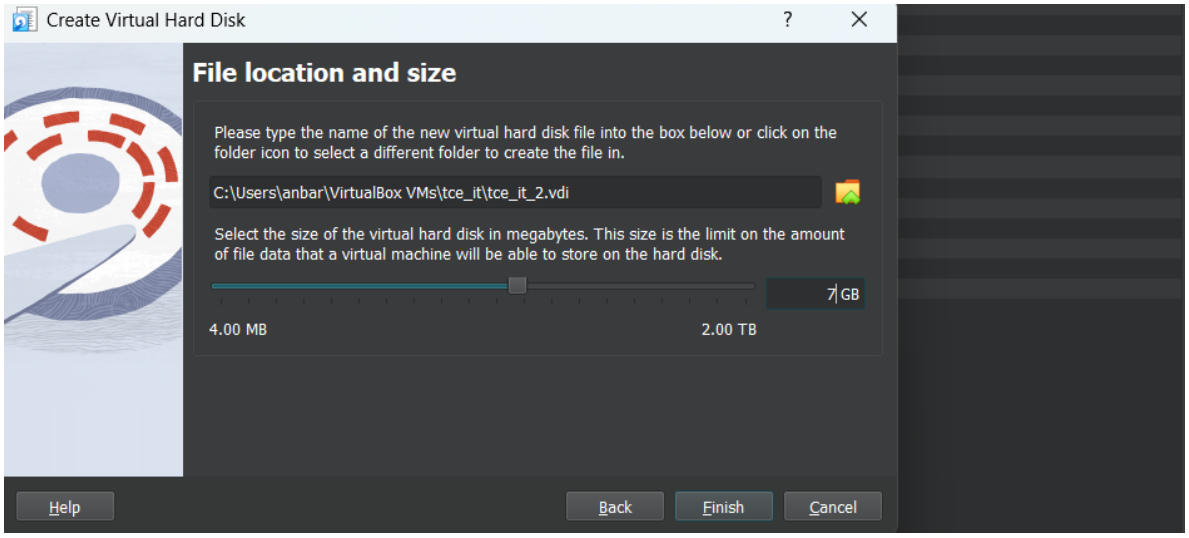
2. **Click the Controller: SATA → Add Hard Disk → Create New Disk.**

3. **Choose VDI (VirtualBox Disk Image), Dynamically allocated,**

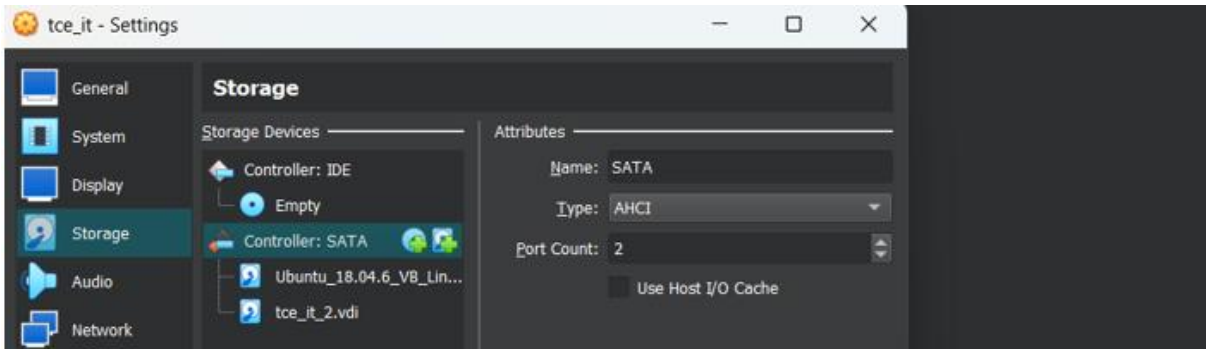




size = 7 GB.



4. **Click Finish.**



Step 2: Start the VM and Login as Root

```
ubuntu@ubuntu1804:~$ sudo -s
```

Step 3: Check Existing Disks and LVM

List all disks

fdisk -l

```
root@ubuntu1804:~# fdisk -l
Disk /dev/loop0: 2.5 MiB, 2621440 bytes, 5120 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/loop1: 65.1 MiB, 68259840 bytes, 133320 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/loop2: 61.8 MiB, 64770048 bytes, 126504 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sda: 512 GiB, 549755813888 bytes, 1073741824 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x67451bf8

Device     Boot Start        End    Sectors  Size Id Type
/dev/sda1  *     2048 1073739775 1073737728  512G 8e Linux LVM


Disk /dev/sdb: 7 GiB, 7516192768 bytes, 14680064 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

List physical volumes

pvdisplay

```
root@ubuntu1804:~# pvdisplay
  --- Physical volume ---
  PV Name               /dev/sda1
  VG Name               ubuntu-vg
  PV Size               <512.00 GiB / not usable 2.00 MiB
  Allocatable           yes (but full)
  PE Size               4.00 MiB
  Total PE              131071
  Free PE               0
  Allocated PE          131071
  PV UUID               3F4Ims-mipI-J39Z-Lt6e-d2dv-a2Fz-wUzCld
```

List volume groups

vgdisplay

```
root@ubuntu1804:~# vgdisplay
  --- Volume group ---
  VG Name               ubuntu-vg
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  3
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                2
  Open LV               2
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               <512.00 GiB
  PE Size               4.00 MiB
  Total PE              131071
  Alloc PE / Size       131071 / <512.00 GiB
  Free  PE / Size       0 / 0
  VG UUID               3Mvbsr-MdcR-wlWG-r4GS-nouT-AMmM-33sdPM
```

List logical volumes

lvdisplay

```
root@ubuntu1804:~# lvdisplay
  --- Logical volume ---
  LV Path                /dev/ubuntu-vg/root
  LV Name                root
  VG Name                ubuntu-vg
  LV UUID                gDvWb8-E4oV-ogjV-UDT7-wx2V-gbCk-Z0FRjv
  LV Write Access        read/write
  LV Creation host, time ubuntu, 2022-03-07 23:20:07 -0500
  LV Status              available
  # open                 1
  LV Size                <511.04 GiB
  Current LE             130826
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:0
```

```
  --- Logical volume ---
  LV Path                /dev/ubuntu-vg/swap_1
  LV Name                swap_1
  VG Name                ubuntu-vg
  LV UUID                PL8auK-uz9c-Wzlf-mCYt-Xcnl-SLki-7qcjWu
  LV Write Access        read/write
  LV Creation host, time ubuntu, 2022-03-07 23:20:07 -0500
  LV Status              available
  # open                 2
  LV Size                980.00 MiB
  Current LE             245
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     256
  Block device           253:1
```

Step 4: Create Physical Volume

pvcreate /dev/sdb

```
root@ubuntu1804:~# pvcreate /dev/sdb
  Physical volume "/dev/sdb" successfully created.
```

Initializes the new disk for LVM.

Step 5: Extend the Volume Group

vgextend ubuntu-vg /dev/sdb

```
root@ubuntu1804:~# vgextend ubuntu-vg /dev/sdb
  Volume group "ubuntu-vg" successfully extended
```

Replace cloud-vg with your existing Volume Group name.

This adds the new PV to your VG.

Step 6: Extend the Logical Volume (Root)

lvresize --size +3G /dev/ubuntu-vg/root

- **+3G → amount of space to add**
- **/dev/cloud-vg/root → your logical volume**

```
root@ubuntu1804:~# lvresize --size +3G /dev/ubuntu-vg/root
  Size of logical volume ubuntu-vg/root changed from <511.04 GiB (130826 extents) to <514.04 GiB
(131594 extents).
  Logical volume ubuntu-vg/root successfully resized.
```

Alternative:

lvextend -L +3G /dev/ubuntu-vg/root

Step 7: Check Current Disk Usage

df -h

Shows current available space on /.

```
root@ubuntu1804:~# df -h
Filesystem                  Size  Used Avail Use% Mounted on
udev                        969M     0  969M   0% /dev
tmpfs                       199M  1.6M  198M   1% /run
/dev/mapper/ubuntu--vg-root  503G  5.9G  471G   2% /
tmpfs                       994M     0  994M   0% /dev/shm
tmpfs                       5.0M  4.0K  5.0M   1% /run/lock
tmpfs                       994M     0  994M   0% /sys/fs/cgroup
/dev/loop0                  2.5M  2.5M     0 100% /snap/gnome-system-monitor/163
/dev/loop3                  768K  768K     0 100% /snap/gnome-characters/741
/dev/loop1                   66M   66M     0 100% /snap/gtk-common-themes/1515
```

```
/dev/loop2                   62M   62M     0 100% /snap/core20/1081
/dev/loop4                  219M  219M     0 100% /snap/gnome-3-34-1804/77
/dev/loop6                  2.5M  2.5M     0 100% /snap/gnome-calculator/884
/dev/loop7                  640K  640K     0 100% /snap/gnome-logs/106
/dev/loop8                   56M   56M     0 100% /snap/core18/2284
/dev/loop9                  242M  242M     0 100% /snap/gnome-3-38-2004/70
/dev/loop10                 219M  219M     0 100% /snap/gnome-3-34-1804/72
/dev/loop12                 128K  128K     0 100% /snap/bare/5
/dev/loop11                 249M  249M     0 100% /snap/gnome-3-38-2004/99
/dev/loop13                 2.7M  2.7M     0 100% /snap/gnome-system-monitor/174
/dev/loop14                 2.7M  2.7M     0 100% /snap/gnome-calculator/920
/dev/loop15                  44M   44M     0 100% /snap/snapd/14978
/dev/loop16                  66M   66M     0 100% /snap/gtk-common-themes/1519
/dev/loop17                  62M   62M     0 100% /snap/core20/1361
/dev/loop18                 768K  768K     0 100% /snap/gnome-characters/726
tmpfs                       199M   28K  199M   1% /run/user/121
tmpfs                       199M   40K  199M   1% /run/user/1000
/dev/loop19                  56M   56M     0 100% /snap/core18/2952
/dev/loop20                  51M   51M     0 100% /snap/snapd/25202
```

Step 8: Resize the Filesystem

resize2fs /dev/mapper/cloud--vg-root

```
root@ubuntu1804:~# resize2fs /dev/mapper/ubuntu--vg-root
resize2fs 1.44.1 (24-Mar-2018)
Filesystem at /dev/mapper/ubuntu--vg-root is mounted on /; on-line resizing required
old_desc_blocks = 64, new_desc_blocks = 65
The filesystem on /dev/mapper/ubuntu--vg-root is now 134752256 (4k) blocks long.
```

Resizes the filesystem to use the newly added space.

Step 9: Verify Disk Space

df -h

```
root@ubuntu1804:~# df -h
Filesystem                  Size  Used Avail Use% Mounted on
udev                        969M     0  969M   0% /dev
tmpfs                       199M  1.6M  198M   1% /run
/dev/mapper/ubuntu--vg-root  505G  5.9G  474G   2% /
tmpfs                       994M     0  994M   0% /dev/shm
tmpfs                       5.0M  4.0K  5.0M   1% /run/lock
tmpfs                       994M     0  994M   0% /sys/fs/cgroup
/dev/loop0                  2.5M  2.5M     0 100% /snap/gnome-system-monitor/163
/dev/loop3                  768K  768K     0 100% /snap/gnome-characters/741
/dev/loop1                   66M   66M     0 100% /snap/gtk-common-themes/1515
/dev/loop2                   62M   62M     0 100% /snap/core20/1081
/dev/loop4                  219M  219M     0 100% /snap/gnome-3-34-1804/77
/dev/loop6                  2.5M  2.5M     0 100% /snap/gnome-calculator/884
/dev/loop7                  640K  640K     0 100% /snap/gnome-logs/106
/dev/loop8                   56M   56M     0 100% /snap/core18/2284
/dev/loop9                  242M  242M     0 100% /snap/gnome-3-38-2004/70
/dev/loop10                 219M  219M     0 100% /snap/gnome-3-34-1804/72
/dev/loop12                 128K  128K     0 100% /snap/bare/5
/dev/loop11                 249M  249M     0 100% /snap/gnome-3-38-2004/99
/dev/loop13                 2.7M  2.7M     0 100% /snap/gnome-system-monitor/174
/dev/loop14                 2.7M  2.7M     0 100% /snap/gnome-calculator/920
/dev/loop15                  44M   44M     0 100% /snap/snapd/14978
/dev/loop16                  66M   66M     0 100% /snap/gtk-common-themes/1519
/dev/loop17                  62M   62M     0 100% /snap/core20/1361
/dev/loop18                 768K  768K     0 100% /snap/gnome-characters/726
tmpfs                       199M   28K  199M   1% /run/user/121
tmpfs                       199M   40K  199M   1% /run/user/1000
/dev/loop19                  56M   56M     0 100% /snap/core18/2952
/dev/loop20                  51M   51M     0 100% /snap/snapd/25202
root@ubuntu1804:~#
```

Confirm that the root filesystem now has additional 3 GB.

Result:

Thus we have created VHD on windows and restored it on another machine successfully.